Cyclotomic Economic Field Theory with 9 Districts

A Framework that Competes with the Standard Nuclear oliGARCHy

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Abstract

This paper develops a theoretical economy with nine districts governed by the ninth cyclotomic polynomial

$$\Phi_9(x) = x^6 + x^3 + 1.$$

The districts are arranged on a cycle, but the admissible economic field is not merely circular. It is constrained by a primitive ninth-root structure that separates aggregate dynamics, triadic block dynamics, and genuinely nine-district dynamics. The theory combines mathematical economics, finance, political economy, geopolitics, international relations, warfare economics, statistics, algorithms, and artificial intelligence. The core condition is

$$\Phi_9(P)z = 0,$$

where P is the cyclic shift operator and z is the vector of district net surpluses. This yields hidden triadic clearing constraints

$$z_i + z_{i+3} + z_{i+6} = 0, \qquad i = 0, 1, 2,$$

while retaining a six-dimensional primitive economic subspace. The resulting framework provides a compact field-theoretic model of production, trade, prices, finance, security, conflict, policy design, and data-driven estimation in a nine-district economy.

The paper ends with "The End"

Keywords: cyclotomic economics, cyclotomic economic fields, nine-district model, geopolitics, warfare economics, macro-finance, artificial intelligence, statistical mechanics.

Contents

List of Figures

List of Tables

1 Introduction

Consider an economy divided into nine districts,

$$i \in \mathbb{Z}_9 = \{0, 1, 2, \dots, 8\}.$$

The districts form a circular economic geography. Goods, capital, information, risk, political authority, and security externalities move along this cycle. The mathematical object governing the economy is the ninth cyclotomic polynomial

$$\Phi_9(x) = x^6 + x^3 + 1.$$

If

$$\zeta = \exp(2\pi \mathrm{i}/9),$$

then ζ is a primitive ninth root of unity and

$$\Phi_9(\zeta) = \zeta^6 + \zeta^3 + 1 = 0.$$

The interpretation is that a truly nine-district economy cannot be reduced to a one-district aggregate economy or to three independent three-district economies. The primitive ninth roots generate a six-dimensional space of genuine nine-district fluctuation. This paper calls that space the *primitive cyclotomic field* of the economy.

The main contributions are as follows.

1. A nine-district production and trade economy is constructed on the cyclic group $\mathbb{Z}_9$.
2. The cyclotomic constraint $\Phi_9(P)z = 0$ is interpreted as a structural clearing rule.
3. No-arbitrage prices are characterized as triadic price anchors.
4. A dynamic field model is introduced using circulant operators.
5. Finance, geopolitics, security, and warfare economics are embedded into the same district field.
6. Statistical, econometric, and AI/ML procedures are proposed for estimating primitive economic structure.

2 Algebraic Preliminaries

2.1 The Ninth Cyclotomic Polynomial

The ninth roots of unity are the solutions of

$$x^9 - 1 = 0.$$

They decompose as

$$x^9 - 1 = (x - 1)(x^2 + x + 1)(x^6 + x^3 + 1).$$

Thus

$$\Phi_9(x) = x^6 + x^3 + 1.$$

The primitive exponents modulo 9 are

$$\mathcal{K}_9 = \{1, 2, 4, 5, 7, 8\}.$$

These are precisely the integers m satisfying

$$\gcd(m, 9) = 1.$$

2.2 The Cyclic Shift Operator

Let $x = (x_0, \dots, x_8)^\top \in \mathbb{R}^9$. Define the cyclic shift operator P by

$$(Px)_i = x_{i-1},$$

where indices are taken modulo 9. Then

$$P^9 = I.$$

The cyclotomic operator is

$$\Phi_9(P) = P^6 + P^3 + I.$$

For any vector $x \in \mathbb{R}^9$,

$$(\Phi_9(P)x)_i = x_i + x_{i-3} + x_{i-6}.$$

Equivalently, after reindexing,

$$x_i + x_{i+3} + x_{i+6}.$$

Definition 1 (Primitive Cyclotomic Field)**.** *The primitive cyclotomic field is the subspace*

$$\mathcal{H}_9 = \operatorname{Ker} \Phi_9(P) = \{x \in \mathbb{R}^9 : x_i + x_{i+3} + x_{i+6} = 0,\ i = 0, 1, 2\}.$$

Since the three restrictions are independent,

$$\dim \mathcal{H}_9 = 6.$$

2.3 Projection onto the Primitive Field

Define

$$\Pi_9 = I - \frac{1}{3}(I + P^3 + P^6).$$

Then

$$(\Pi_9 x)_i = x_i - \frac{x_i + x_{i+3} + x_{i+6}}{3}.$$

Therefore Π_9 removes the triadic mean inside each of the three hidden triads.

Proposition 1 (Projection)**.** *The matrix* Π_9 *is an orthogonal projection onto* $\mathcal{H}_9$*.*

Proof. First,

$$\Pi_9^2 = \left(I - \frac{1}{3}(I + P^3 + P^6)\right)^2.$$

Since $P^9 = I$, one has

$$(I + P^3 + P^6)^2 = 3(I + P^3 + P^6).$$

Therefore

$$\Pi_9^2 = I - \frac{2}{3}(I + P^3 + P^6) + \frac{1}{9}3(I + P^3 + P^6) = I - \frac{1}{3}(I + P^3 + P^6) = \Pi_9.$$

Because P is orthogonal and $P^3 + P^6$ is symmetric after taking transposes modulo 9, Π_9 is symmetric. Hence Π_9 is an orthogonal projection. Its image consists exactly of vectors whose triadic means vanish, namely $\mathcal{H}_9$. $\square$

3 District Geography and Economic Roles

The economy has nine districts placed on a cycle.

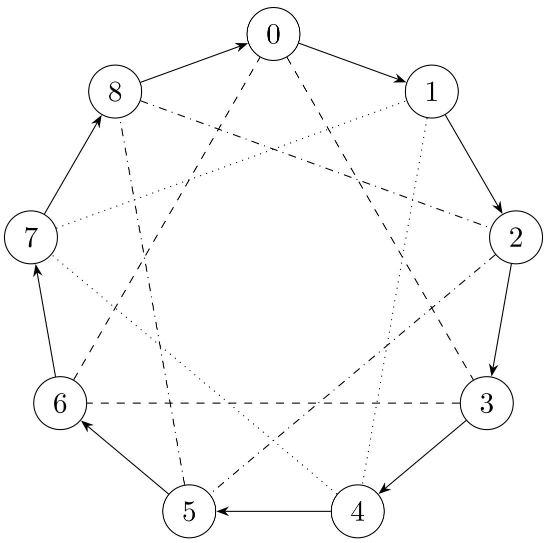
$$0 \to 1 \to 2 \to 3 \to 4 \to 5 \to 6 \to 7 \to 8 \to 0.$$

The hidden triads are

$$\mathcal{T}_0 = \{0, 3, 6\}, \qquad \mathcal{T}_1 = \{1, 4, 7\}, \qquad \mathcal{T}_2 = \{2, 5, 8\}.$$

Table 1: Nine-district economic interpretation

District	Main role	Hidden triad
0	Treasury, clearing, capital markets	Asset triad
1	Agriculture, food, biological inputs	Goods triad
2	Port, foreign exchange, external trade	Risk-information triad
3	Energy, utilities, infrastructure	Asset triad
4	Manufacturing, industrial processing	Goods triad
5	Universities, AI, data, research	Risk-information triad
6	Land, housing, minerals, reserves	Asset triad
7	Logistics, storage, transportation	Goods triad
8	Insurance, defense, resilience	Risk-information triad



visible 9-cycle with hidden triads

Figure 1: The nine-district cycle and the three hidden triads.

4 The Economic Field

4.1 State Variables

Each district i has state variables

$$x_i(t) = \big(y_i(t), c_i(t), k_i(t), \ell_i(t), G_i(t), b_i(t), s_i(t), m_i(t)\big),$$

where

$y_i(t)$	output,
$c_i(t)$	consumption,
$k_i(t)$	capital,
$\ell_i(t)$	labor,
$G_i(t)$	public good,
$b_i(t)$	net financial position,
$s_i(t)$	security stock,
$m_i(t)$	military or coercive capacity index.

The district output vector is

$$y(t) = (y_0(t), \dots, y_8(t))^\top.$$

The net surplus vector is

$$z_i(t) = y_i(t) - c_i(t) - I_i(t) - G_i(t) - D_i(t),$$

where $I_i(t)$ is investment and $D_i(t)$ is defense or resilience expenditure.

4.2 Cyclotomic Clearing

Axiom 1 (Cyclotomic Clearing). *At each date t, feasible net surpluses satisfy*

$$\Phi_9(P)z(t) = 0.$$

Equivalently,

$$z_i(t) + z_{i+3}(t) + z_{i+6}(t) = 0, \qquad i = 0, 1, 2.$$

Thus every hidden triad clears internally:

$$z_0 + z_3 + z_6 = 0,$$

$$z_1 + z_4 + z_7 = 0,$$

$$z_2 + z_5 + z_8 = 0.$$

Definition 2 (Primitive Circulation). *A net surplus vector z is a primitive circulation if*

$$z \in \mathcal{H}_9.$$

Primitive circulation means that the economy has active interdistrict motion without reducibility to either one aggregate market or three isolated bloc markets.

5 Households, Firms, and Production

5.1 Households

District i has a representative household with utility

$$U_i(c_i, \ell_i, G_i, s_i) = \log c_i - \frac{\chi_i}{2}\ell_i^2 + \theta_i \log G_i + \psi_i \log s_i,$$

where

$$\chi_i, \theta_i, \psi_i > 0.$$

The household budget constraint is

$$p_i c_i + q_i a_i' \leq w_i \ell_i + r_i a_i + \tau_i + \pi_i,$$

where p_i is the consumption price, q_i is the asset price, a_i is asset holdings, w_i is the wage, r_i is the gross return on assets, τ_i is net transfers, and π_i is distributed firm profit.

The first-order conditions imply

$$\frac{1}{c_i} = \lambda_i p_i,$$

$$\chi_i \ell_i = \lambda_i w_i,$$

and therefore

$$\frac{w_i}{p_i} = \chi_i c_i \ell_i.$$

5.2 Firms

District i has production technology

$$y_i = A_i e^{-\omega_i} k_i^{\alpha} \ell_i^{1-\alpha}, \qquad 0 < \alpha < 1.$$

Here $\omega_i \geq 0$ is the local disruption index from insecurity, political conflict, sanctions, blockade risk, or logistics failure.

Profit is

$$\Pi_i = p_i y_i - w_i \ell_i - r_i k_i.$$

The first-order conditions are

$$w_i = p_i (1-\alpha) A_i e^{-\omega_i} k_i^{\alpha} \ell_i^{-\alpha},$$

$$r_i = p_i \alpha A_i e^{-\omega_i} k_i^{\alpha-1} \ell_i^{1-\alpha}.$$

6 Cyclotomic Productivity

Productivity is decomposed into primitive ninth harmonics:

$$\log A_i = a_0 + \sum_{m \in \{1,2,4\}} \left[a_m \cos\left(\frac{2\pi m i}{9}\right) + b_m \sin\left(\frac{2\pi m i}{9}\right) \right].$$

The exponents $1, 2, 4, 5, 7, 8$ are the primitive ninth-root modes. Hence productivity fluctuations are genuinely nine-district and not reducible to triadic averages.

Definition 3 (Cyclotomic Productivity). *A productivity vector $A \in \mathbb{R}^9_+$ is cyclotomic if*

$$\Pi_9 \log A = \log A - \frac{1}{3}(I + P^3 + P^6) \log A$$

contains the economically relevant regional productivity information.

7 Prices and No-Arbitrage

Let $p = (p_0, \ldots, p_8)^\top$ be the vector of goods prices.

Definition 4 (Cyclotomic No-Arbitrage). *A price vector p is cyclotomic no-arbitrage if*

$$p^\top z = 0$$

for every feasible primitive circulation $z \in \mathcal{H}_9$.

Proposition 2 (Triadic Price Anchors)**.** *A price vector satisfies cyclotomic no-arbitrage if and only if*

$$p_0 = p_3 = p_6, \qquad p_1 = p_4 = p_7, \qquad p_2 = p_5 = p_8.$$

Proof. Suppose $p^\top z = 0$ for every $z \in \mathcal{H}_9$. For districts 0 and 3, choose

$$z = e_0 - e_3.$$

This vector belongs to $\mathcal{H}_9$, because its sum within the triad $\{0, 3, 6\}$ is zero and all other triad sums are zero. Then

$$0 = p^\top z = p_0 - p_3.$$

Thus $p_0 = p_3$. Similarly, using $e_3 - e_6$, one obtains $p_3 = p_6$. The same argument applies to $\{1, 4, 7\}$ and $\{2, 5, 8\}$. Conversely, if prices are constant within each hidden triad, then for any $z \in \mathcal{H}_9$,

$$p^\top z = p_0(z_0 + z_3 + z_6) + p_1(z_1 + z_4 + z_7) + p_2(z_2 + z_5 + z_8) = 0.$$

□

Thus the economy has nine districts but three fundamental price anchors:

$$p^A = p_0 = p_3 = p_6,$$

$$p^B = p_1 = p_4 = p_7,$$

$$p^C = p_2 = p_5 = p_8.$$

8 Competitive Equilibrium

Definition 5 (Cyclotomic Competitive Equilibrium)**.** *A cyclotomic competitive equilibrium is a collection*

$$\{c_i, \ell_i, k_i, y_i, G_i, D_i, z_i, p_i, w_i, r_i\}_{i=0}^{8}$$

such that:

1. *households maximize utility subject to budget constraints;*
2. *firms maximize profits subject to production technology;*
3. *public goods and security stocks are feasible;*
4. *goods markets clear through the cyclotomic condition*

$$\Phi_9(P)z = 0;$$

5. *prices satisfy triadic no-arbitrage;*
6. *labor, capital, fiscal, and financial markets clear.*

Assumption 1 (Regularity)**.** *Preferences are continuous, strictly concave, and locally nonsatiated. Production sets are closed, convex after standard profit-set transformation, and satisfy free disposal. Endowments are strictly positive. Security losses ω_i are bounded.*

Theorem 1 (Existence)**.** *Under the regularity assumption, a cyclotomic competitive equilibrium exists.*

Proof. The model is a finite-dimensional Arrow–Debreu economy with additional linear feasibility restrictions. The household choice sets are convex and compact after imposing aggregate resource bounds. Preferences are continuous and strictly concave. Production sets are closed and convex after the usual convexification argument. The cyclotomic clearing restriction

$$\Phi_9(P)z = 0$$

is a closed linear constraint, so it preserves convexity and compactness of the feasible allocation set. The standard fixed-point equilibrium existence argument therefore applies on the constrained feasible set. Since prices are defined up to normalization and the no-arbitrage subspace is nonempty, there exists a supporting price vector satisfying the triadic anchor conditions. □

9 Finance and Asset Pricing

Let R_i denote the gross return on a district-specific asset. Decompose returns as

$$R_i = R_f + \mu_i + \varepsilon_i,$$

where R_f is the common risk-free return, μ_i is a risk premium, and ε_i is an idiosyncratic innovation.

The primitive risk premium is

$$\mu^{\text{prim}} = \Pi_9 \mu.$$

A representative stochastic discount factor M prices assets:

$$1 = \mathbb{E}[MR_i], \qquad i = 0, \ldots, 8.$$

The excess-return condition is

$$\mathbb{E}[R_i - R_f] = -\frac{\operatorname{Cov}(M, R_i)}{\mathbb{E}[M]}.$$

The cyclotomic decomposition gives

$$\mathbb{E}[\Pi_9(R - R_f\mathbf{1})] = -\frac{1}{\mathbb{E}[M]}\Pi_9 \operatorname{Cov}(M, R).$$

Hence primitive excess returns are compensation for exposure to primitive ninth-district risk.

Definition 6 (Cyclotomic Risk Factor)**.** *The cyclotomic risk factor is*

$$F_9(t) = v^\top \Pi_9 R(t),$$

where $v \in \mathcal{H}_9$ is a normalized primitive portfolio weight vector.

A district portfolio is cyclotomic-neutral if

$$v^\top \Pi_9 R = 0.$$

10 Political Economy and Public Finance

Let g_i denote fiscal expenditure and T_i tax revenue in district i. The government budget constraint is

$$\sum_{i=0}^{8} p_i g_i + B' = \sum_{i=0}^{8} T_i + RB.$$

A cyclotomic fiscal rule penalizes nonprimitive distortions:

$$\mathcal{L}_{\text{fiscal}}(z) = \frac{\kappa}{2}\|\Phi_9(P)z\|^2.$$

When $\Phi_9(P)z = 0$, the circulation is primitive and receives no penalty. When a trade pattern collapses into a degenerate aggregate or bloc imbalance, the penalty is positive.

The social planner solves

$$\max_{\{c_i,\ell_i,G_i,D_i,k_i'\}} \sum_{i=0}^{8} \lambda_i \left[\log c_i - \frac{\chi_i}{2}\ell_i^2 + \theta_i \log G_i + \psi_i \log s_i\right] - \frac{\kappa}{2}\|\Phi_9(P)z\|^2$$

subject to production, accumulation, and security constraints.

Capital accumulation is

$$k_i' = (1-\delta)k_i + I_i.$$

Security accumulation is

$$s_i' = (1-\delta_s)s_i + \eta_i D_i^{\rho}, \qquad 0 < \rho < 1.$$

11 Geopolitics and International Relations

Suppose an external actor applies a pressure vector

$$h = (h_0, \ldots, h_8)^{\top},$$

where h_i may represent sanctions, border frictions, diplomatic pressure, treaty support, foreign investment, or external security assistance.

The primitive geopolitical pressure is

$$h^{\text{prim}} = \Pi_9 h.$$

The aggregate pressure is

$$h^{\text{agg}} = \frac{1}{9}\mathbf{1}\mathbf{1}^{\top}h.$$

The triadic pressure is

$$h^{\text{triad}} = \frac{1}{3}(I + P^3 + P^6)h - h^{\text{agg}}.$$

Hence

$$h = h^{\text{agg}} + h^{\text{triad}} + h^{\text{prim}}.$$

Proposition 3 (Primitive Geopolitical Shocks)**.** *A geopolitical shock h is invisible to aggregate accounting and triadic bloc accounting if and only if*

$$h = \Pi_9 h.$$

Proof. The aggregate and triadic components are removed exactly by Π_9. Therefore h has no aggregate or triadic component precisely when $h \in \mathrm{Im}(\Pi_9) = \mathcal{H}_9$. □

This implies that an external power can perturb the nine-district economy without changing total output or the apparent balance of the three hidden triads. Such pressure is difficult to detect using aggregate statistics.

12 Warfare Economics and Security

The model uses a reduced-form, non-operational representation of conflict. Let $m_i \geq 0$ be a district's coercive capacity index and let $\bar{m}_i \geq 0$ be rival pressure near district i. Define a contest-success function

$$\pi_i^C = \frac{m_i^\eta}{m_i^\eta + \bar{m}_i^\eta}, \qquad \eta > 0.$$

The disruption index is

$$\omega_i = \bar{\omega}_i(1 - \pi_i^C) + \xi_i,$$

where ξ_i is exogenous insecurity.

Output is therefore

$$y_i = A_i e^{-\omega_i} k_i^\alpha \ell_i^{1-\alpha}.$$

The marginal economic value of security is

$$\frac{\partial y_i}{\partial m_i} = -y_i \frac{\partial \omega_i}{\partial m_i}.$$

Since

$$\frac{\partial \pi_i^C}{\partial m_i} = \frac{\eta m_i^{\eta-1} \bar{m}_i^\eta}{(m_i^\eta + \bar{m}_i^\eta)^2},$$

we have

$$\frac{\partial \omega_i}{\partial m_i} = -\bar{\omega}_i \frac{\eta m_i^{\eta-1} \bar{m}_i^\eta}{(m_i^\eta + \bar{m}_i^\eta)^2}.$$

Thus

$$\frac{\partial y_i}{\partial m_i} = y_i \bar{\omega}_i \frac{\eta m_i^{\eta-1} \bar{m}_i^\eta}{(m_i^\eta + \bar{m}_i^\eta)^2}.$$

Definition 7 (Cyclotomic Security Externality)**.** *The primitive security externality is*

$$\Omega^{\text{prim}} = \Pi_9 \omega.$$

Security policy is cyclotomic-stabilizing if it minimizes

$$\|\Pi_9 \omega\|^2$$

subject to a budget constraint.

13 Dynamic Field Theory

Let $x_t \in \mathbb{R}^9$ be the vector of output deviations from steady state:

$$x_t = y_t - \bar{y}.$$

Define the dynamic law

$$x_{t+1} = Bx_t + \Gamma u_t + \varepsilon_t,$$

where u_t is a vector of policy instruments and ε_t is a shock.

A symmetric primitive circulant operator is

$$B = b_1(P + P^{-1}) + b_2(P^2 + P^{-2}) + b_4(P^4 + P^{-4}).$$

Since P is diagonalized by the discrete Fourier basis, the eigenvalues of B are

$$\lambda_m = 2b_1 \cos\left(\frac{2\pi m}{9}\right) + 2b_2 \cos\left(\frac{4\pi m}{9}\right) + 2b_4 \cos\left(\frac{8\pi m}{9}\right),$$

for $m = 0, \ldots, 8$.

The primitive modes are $m \in \mathcal{K}_9 = \{1, 2, 4, 5, 7, 8\}$.

Theorem 2 (Primitive Stability)**.** *The primitive cyclotomic economy is locally stable if and only if*

$$|\lambda_m| < 1$$

for every

$$m \in \{1, 2, 4, 5, 7, 8\}.$$

Proof. The operator B is circulant and therefore diagonalizable by the discrete Fourier matrix. Each Fourier mode evolves independently as

$$\hat{x}_{m,t+1} = \lambda_m \hat{x}_{m,t}.$$

The primitive subspace is spanned by the modes $m \in \{1, 2, 4, 5, 7, 8\}$. Stability on that subspace is equivalent to convergence of every primitive scalar recursion, which holds precisely when $|\lambda_m| < 1$ for every primitive m. $\square$

14 Statistical Identification

Suppose we observe a panel

$$\{x_i(t) : i = 0, \dots, 8,\ t = 1, \dots, T\}.$$

The discrete Fourier transform is

$$\hat{x}_m(t) = \frac{1}{\sqrt{9}} \sum_{j=0}^{8} x_j(t)\zeta^{-mj}, \qquad m = 0, \dots, 8.$$

Define primitive spectral power

$$S_{\text{prim}} = \sum_{m \in \mathcal{K}_9} \frac{1}{T} \sum_{t=1}^{T} |\hat{x}_m(t)|^2.$$

Define nonprimitive spectral power

$$S_{\text{non}} = \sum_{m \in \{0,3,6\}} \frac{1}{T} \sum_{t=1}^{T} |\hat{x}_m(t)|^2.$$

The cyclotomic intensity statistic is

$$C_9 = \frac{S_{\text{prim}}}{S_{\text{prim}} + S_{\text{non}}}.$$

If C_9 is close to one, the economy is dominated by primitive nine-district variation. If C_9 is close to zero, the economy is mostly aggregate or triadic.

14.1 Regression Model

A reduced-form econometric model is

$$x_{t+1} = B_\theta x_t + \Gamma u_t + \varepsilon_t,$$

where

$$B_\theta = b_1(P + P^{-1}) + b_2(P^2 + P^{-2}) + b_4(P^4 + P^{-4}).$$

The least-squares estimator solves

$$(\hat{b}_1, \hat{b}_2, \hat{b}_4, \hat{\Gamma}) = \arg\max_{b_1, b_2, b_4, \Gamma} - \sum_{t=1}^{T-1} \|x_{t+1} - B_\theta x_t - \Gamma u_t\|^2.$$

14.2 Hypothesis Testing

A useful null hypothesis is

$$H_0 : S_{\text{prim}} = 0.$$

The alternative is

$$H_1 : S_{\text{prim}} > 0.$$

A simple test statistic is

$$\mathcal{T}_9 = T \frac{S_{\text{prim}}}{S_{\text{non}} + \epsilon},$$

where $\epsilon > 0$ is a small regularization constant.

15 Algorithms

Table 2: Algorithm 1: Cyclotomic field decomposition

Step	Operation
1	Input district vector $x \in \mathbb{R}^9$.
2	Compute the triadic average $a = (I + P^3 + P^6)x/3$.
3	Compute the primitive component $x^{\text{prim}} = x - a$.
4	Compute the aggregate component $x^{\text{agg}} = (\mathbf{1}^\top x/9)\mathbf{1}$.
5	Compute the triadic nonaggregate component $x^{\text{triad}} = a - x^{\text{agg}}$.
6	Return $x = x^{\text{agg}} + x^{\text{triad}} + x^{\text{prim}}$.

Table 3: Algorithm 2: Estimation of primitive dynamic coefficients

Step	Operation
1	Input panel data $\{x_t, u_t\}_{t=1}^T$.
2	Construct P, P^2, P^4 and their inverses.
3	Form $B_\theta = b_1(P + P^{-1}) + b_2(P^2 + P^{-2}) + b_4(P^4 + P^{-4})$.
4	Minimize $\sum_t \|x_{t+1} - B_\theta x_t - \Gamma u_t\|^2$.
5	Estimate primitive eigenvalues $\hat{\lambda}_m$ for $m \in \mathcal{K}_9$.
6	Report stability if $\lvert\hat{\lambda}_m\rvert < 1$ for all primitive m.

16 AI and Machine Learning Architecture

A cyclotomic equivariant prediction model should commute with the cyclic shift:

$$f(Px) = Pf(x).$$

A linear equivariant layer has the form

$$f_\Theta(x) = \sum_{r=0}^{8} \theta_r P^r x.$$

A primitive-filtered neural layer is

$$h_{l+1} = \sigma\left(\Pi_9 \sum_{r=0}^{8} \theta_{l,r} P^r h_l + W_l u_t\right),$$

where σ is a nonlinear activation.

The training objective is

$$\min_{\Theta} \sum_{t=1}^{T-1} \|x_{t+1} - f_\Theta(x_t, u_t)\|^2 + \lambda \sum_{t=1}^{T} \|\Phi_9(P) f_\Theta(x_t, u_t)\|^2 .$$

The second term penalizes predictions that violate cyclotomic feasibility.

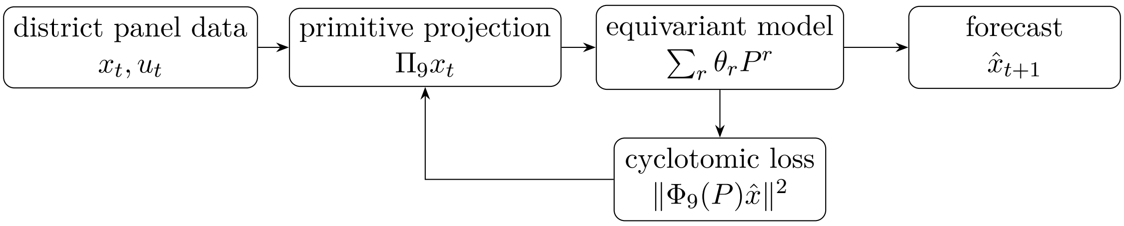


Figure 2: Primitive-filtered AI/ML architecture for nine-district forecasting.

17 Policy Instruments

Table 4: Policy instruments in the nine-district cyclotomic economy

Policy instrument	Mathematical object	Primary target
Fiscal transfers	$\tau \in \mathbb{R}^9$	Consumption smoothing
Infrastructure spending	$G \in \mathbb{R}^9_+$	Productivity and connectivity
Security spending	$D \in \mathbb{R}^9_+$	Disruption reduction
Credit allocation	$b \in \mathbb{R}^9$	Financial stabilization
Trade regulation	$z \in \mathcal{H}_9$	Primitive circulation
Sanctions response	$\Pi_9 h$	Geopolitical resilience
AI targeting	$f_\Theta(Px) = P f_\Theta(x)$	Equivariant prediction

A policy vector u is primitive-neutral if

$$\Pi_9 u = 0.$$

It is primitive-active if

$$\Pi_9 u \neq 0.$$

Primitive-active policy affects genuine nine-district structure, while primitive-neutral policy affects only aggregates or triadic blocs.

18 A General Welfare Functional

The welfare functional is

$$W = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\sum_{i=0}^{8} \lambda_i \Big(\log c_i(t) - \frac{\chi_i}{2} \ell_i(t)^2 + \theta_i \log G_i(t) + \psi_i \log s_i(t) \Big) - \frac{\kappa}{2} \|\Phi_9(P) z(t)\|^2 - \frac{\nu}{2} \|\Pi_9 \omega(t)\|^2 \right].$$

The term

$$\|\Phi_9(P) z(t)\|^2$$

penalizes violations of cyclotomic clearing.

The term

$$\|\Pi_9\omega(t)\|^2$$

penalizes primitive security distortions.

The planner's first-order condition for a generic policy u is

$$\mathbb{E}_t\left[\frac{\partial W}{\partial u}\right] = 0.$$

For security expenditure D_i, the symbolic first-order condition is

$$\lambda_i\psi_i\frac{1}{s_i}\frac{\partial s_i}{\partial D_i} + \lambda_i\frac{1}{c_i}\frac{\partial c_i}{\partial D_i} - \nu\left(\Pi_9\omega\right)^\top \Pi_9\frac{\partial \omega}{\partial D_i} = 0.$$

This condition balances local security benefits, consumption opportunity costs, and primitive systemic security externalities.

19 Core Mathematical Characterization

Theorem 3 (Cyclotomic Economic Field Characterization). *A nine-district economy is a cyclotomic economic field if its feasible surplus vectors satisfy*

$$z \in \operatorname{Ker}\Phi_9(P).$$

Equivalently, for each hidden triad,

$$z_i + z_{i+3} + z_{i+6} = 0, \qquad i = 0, 1, 2.$$

The feasible surplus space has dimension 6 *and is spanned by the primitive ninth-root modes.*

Proof. The equality

$$\Phi_9(P) = I + P^3 + P^6$$

implies

$$\Phi_9(P)z = 0$$

if and only if

$$z_i + z_{i+3} + z_{i+6} = 0$$

for each residue class modulo 3. These are three independent linear restrictions in $\mathbb{R}^9$, so the solution space has dimension $9 - 3 = 6$. Since P is diagonalized by the ninth roots of unity and Φ_9 vanishes exactly on primitive ninth roots, the kernel is spanned by the primitive modes $m \in \{1, 2, 4, 5, 7, 8\}$. □

Corollary 1 (Irreducibility). *The cyclotomic economy cannot be fully represented by a one-district aggregate model or by a three-bloc model.*

Proof. The aggregate and triadic modes correspond to Fourier indices $m \in \{0, 3, 6\}$. The primitive field is spanned by the complementary indices $\{1, 2, 4, 5, 7, 8\}$. Therefore aggregate and triadic representations omit the entire primitive subspace. □

20 Discussion

The ninth cyclotomic polynomial

$$\Phi_9(x) = x^6 + x^3 + 1$$

generates an economic architecture with nine visible districts, three hidden triads, and six primitive degrees of freedom. The visible geography is a ring. The hidden accounting system is triadic. The true field dynamics are primitive and six-dimensional.

The theory has several interpretations.

1. In economics, it is a constrained general equilibrium model with nonstandard regional clearing.
2. In finance, it is a primitive-factor asset-pricing model.
3. In political science, it is a model of regional authority and fiscal balancing.
4. In geopolitics, it separates aggregate, bloc-level, and primitive pressure.
5. In warfare economics, it models insecurity as a production-destroying field.
6. In statistics, it gives a spectral decomposition for detecting hidden regional structure.
7. In AI/ML, it motivates cyclically equivariant and cyclotomically constrained learning systems.

21 Conclusion

This paper constructed a nine-district economy governed by the ninth cyclotomic polynomial. The central equation is

$$\Phi_9(P)z = 0.$$

Because

$$\Phi_9(P) = I + P^3 + P^6,$$

this condition imposes hidden triadic clearing:

$$z_i + z_{i+3} + z_{i+6} = 0.$$

Yet the remaining feasible space is six-dimensional and primitive. The model therefore describes an economy that is neither aggregate nor merely triadic. It is a field of primitive cyclic interdependence.

The resulting framework unifies production, trade, finance, public policy, geopolitics, security, warfare economics, statistics, and machine learning under a single algebraic principle:

$$\boxed{\text{The structure of the economy is the structure of } \Phi_9.}$$

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Glossary

Cyclotomic polynomial
A polynomial whose roots are primitive roots of unity. In this paper, $\Phi_9(x) = x^6 + x^3 + 1$.

Primitive ninth root
A complex number ζ^m with $\gcd(m, 9) = 1$, where $\zeta = e^{2\pi \mathrm{i}/9}$.

Cyclic shift operator
The operator P satisfying $(Px)_i = x_{i-1}$ and $P^9 = I$.

Primitive cyclotomic field
The six-dimensional subspace $\mathcal{H}_9 = \operatorname{Ker} \Phi_9(P)$.

Hidden triad
One of the sets $\{0, 3, 6\}$, $\{1, 4, 7\}$, or $\{2, 5, 8\}$.

Cyclotomic clearing
The feasibility condition $\Phi_9(P)z = 0$.

Primitive circulation
A net surplus vector satisfying $z_i + z_{i+3} + z_{i+6} = 0$ for $i = 0, 1, 2$.

Triadic price anchor
A no-arbitrage condition requiring equal prices inside each hidden triad.

Primitive geopolitical pressure
The projection $\Pi_9 h$ of an external pressure vector onto the primitive field.

Cyclotomic risk factor
A financial risk factor formed from primitive ninth-district return variation.

Primitive stability
Dynamic stability restricted to Fourier modes $m \in \{1, 2, 4, 5, 7, 8\}$.

Cyclotomic intensity
A statistic measuring the share of total spectral power belonging to primitive ninth modes.

Equivariant AI model
A machine-learning model satisfying $f(Px) = Pf(x)$.

Cyclotomic loss
A training penalty of the form $\|\Phi_9(P)\hat{x}\|^2$, used to discourage structurally infeasible predictions.

The End